

BENJAMIN LAUBACH

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Education

University of Michigan, Ann Arbor Expected Graduation: May 2028

Doctor of Philosophy in Chemical Engineering and Scientific Computing

- National Science Foundation Graduate Research Fellowship (NSF-GRFP)

Master of Science in Engineering – Chemical Engineering, University of Michigan, Ann Arbor Graduation: May 2024

Bachelor of Science – Chemical Engineering with Honors, Michigan State University Graduation: May 2023, GPA: 3.90/4.00

Professional & Research Experience

Google X Jan. 2026 – Jun. 2026

PhD Resident – AI for Chemistry with Dr. John Paul Issa, Thomas Hossler, and Prof. Gabe Gomes

- Architected and deployed an end-to-end MLOps pipeline on Google Cloud Platform (GCP) to execute large-scale machine learning inference on 1M+ partner data points.

Los Alamos National Laboratory Jun. 2025 – Aug. 2025

Graduate Research Intern with Dr. Ying Wai Li and Dr. Christopher Lane

- Integrated complex reward and acceleration methods into a Monte Carlo Tree Search Algorithm for the inverse design and discovery of heavy fermion (superconducting) compounds.

University of Michigan-Ann Arbor May. 2023 – Current

Graduate Research Assistant with Prof. Rebecca Lindsey

- Developed methods for uncertainty quantification method for simulations using machine-learned interatomic potentials (ML-IAP); collaboration with Lawrence Livermore National Laboratory.

University of Wisconsin-Madison Jul. 2021 – May 2023

Undergraduate Research Assistant with Prof. Victor Zavala

- Applied Riemannian Geometry to various case studies in collaboration with Dow Chemical. Determined key process variables affecting multivariate time series batch processes. Collaboration with Argonne National Laboratory.

Dow Chemical Company May 2021 – Aug. 2022

Alternating Term Co-op

- Scaled batch production by 25.8% (adding ~\$800k/yr profit), reduced waste by 40% through a Python cleanout model, led capital investment projects, and identified unsustainable SiH-containing products that saved ~\$10k per trailer.

Michigan State University Aug. 2021 – May 2023

Undergraduate Research Assistant with Prof. John Dorgan

- Utilized classification algorithms to sort plastics based on Near Infrared Spectroscopy (NIR) data.

Michigan State University Jan. 2020 – May 2020

Undergraduate Research Assistant with Prof. Milton Smith

- Focused on iridium-catalyzed C-H borylation chemistry with an emphasis on coupling reactions. Included a prioritization of green chemistry with a focus on optimization.

Awards

- Feb. 2025 – 2nd place Business + Tech Data Download and Datathon Competition
- Apr. 2024 – National Science Foundation Graduate Research Fellowship
- Nov. 2022 – 1st overall in Computing and Process Control, 2022 AIChE Annual Meeting Poster Competition
- Sep. 2022 – Bernard A. & Joan L. Paulson Scholarship
- Sep. {2022, 2021, 2020} – Maurice G. and Sara V. Larian Scholarship
- Sep. {2022, 2020} – David L. and Denise M. Lamp Engineers of Tomorrow Endowment Scholarship
- Jun 2019 – Sergio Marchionne Student Achievement Scholarship
- Jun. 2019 – Pinnacle Award Scholarship
- Jun. 2019 – Booster Award Scholarship
- Jun. 2019 – Credit Union One Scholarship

Publications

- **Laubach B.**, Liu S., Allen K., Riedel Z., Watkins A., Rosa P., Thomas S., Bauer E., Ronning F., Lindsey R., Zhu J., Lane C., Li Y., *Machine learning accelerated monte carlo tree search for inverse design of quantum materials*. In preparation.
- **Laubach B.**, Vita J., Bushick K., Williams L., Lindsey R., Lordi V., *Entropy-guided dataset optimization for machine-learned interatomic potentials*. In preparation.
- Oladipupo A., **Laubach B.**, Almohri S., Lindsey R., *TurboChIMES: Multi-Layered Machine-Learned Interatomic Models for Enhanced Simulation Efficiency*. Submitted.
- Vita J., Stimac J., Williams L., Fan Y., Buschick K., Hamel S., Samanta A., **Laubach B.**, Lindsey R., Lordi V., *Cluster-based Bayesian optimization for application-specific interatomic potential training*. Submitted.
- **Laubach B.**, Lordi, V., Lindsey R., *Cluster-Graph Fingerprinting: A Framework for Quantitative Analysis of Machine-Learned Interatomic Model Training and Simulation Data*. ChemRxiv. 2025; doi:10.26434/chemrxiv-2025-vr0cs
- Smith, A., **Laubach, B.**, Castillo, I. & Zavala, V. M., *Data analysis using Riemannian geometry and applications to chemical engineering*. *Computers & Chemical Engineering* 168, 108023 (2022).

Presentations

2025 University of Michigan Chemical Engineering Graduate Symposium – Oral Presentation **Sep. 2025**

Inverse design of heavy fermion compounds using monte carlo tree search

2025 Conference on Data Analysis – Poster Presentation **Apr. 2025**

A New Configuration Fingerprinting Method for Improved Data Selection and Uncertainty Quantification in Machine Learned Interatomic Models

2024 AIChE Annual Meeting – Oral Presentation **Jun. 2024**

Evaluating Machine-Learned Interatomic Models through Graph-based Descriptor Analysis

2024 Midwest Thermodynamics and Statistical Mechanics Conference – Poster Presentation **Jun. 2024**

Detection Toward trustworthy and robust simulation with machine-learned interatomic models: New strategies for quantifying configuration novelty

2022 AIChE Annual Meeting – Undergraduate Poster Competition **Nov. 2022**

Utilizing the Euler Characteristic for Extreme Event Detection

Leadership Experiences

University of Michigan | VP of {Deal Flow, Non-MBA} – Zell Lurie Commercialization Fund **Sep. 2024 – Current**

- Identified young startups for early-stage investment with check sizes ranging from 50K~150K.

University of Michigan | Consultant – miLEAD Consulting Group **Sep. 2024 – Current**

- Solved complex problems facing partners through team-driven, short-term, research projects.

University of Michigan | President, Vice President – Chemical Engineering Graduate Society **Dec. 2023 – Dec. 2025**

- Facilitate initiatives within the chemical engineering graduate department to support student needs.
- Volunteered for the local Peace Neighborhood Center to promote STEM education.

Michigan State University | Founder – Chemical Engineering Introductory Project **Dec. 2020 – May 2023**

- Created an extracurricular activity to teach prospective engineering students' technical skills through hands-on learning.

Michigan State University | Resident Assistant **Mar. 2020 – May 2023**

- Assisted in supporting an on-campus residential community and have specific responsibility for connecting with students in a particular area. RAs work to establish inclusive communities that foster student academic and social success.

Michigan State University | President – AIChE **Sep. 2019 – May 2023**

- Represented AIChE while networking for chemical engineering companies and professionals.

Michigan State University | Honors College Activities Board **Sep. 2019 – Mar. 2020**

- Collaborated with MSU students to host events for honors college members.

Michigan State University | Fluid Flow and Heat Transfer Grader **Aug. 2022 – May 2023**

- Provided constructive feedback to students for Fluid Flow and Heat Transfer homework.

Volunteering

Michigan State University | *Tower Guard*

Mar. 2020 – Aug. 2021

- Tower Guard is a sophomore student organization dedicated to serving 60 volunteer hours to the Resource Center for Persons with Disabilities (RCPD) at Michigan State University.

Michigan State University | *Honors College Impact* **Aug. 2019**

- Week-long volunteering event hosted by the Honors College to help the greater Lansing community.

Oakland University | *Co-facilitator - OUCares Volunteer* **Oct. 2018 – Jan. 2019**

- Volunteered to teach children with autism computer programming through Lego Robotics.

Relevant Coursework

Machine Learning

- Machine Learning, Graduate **Fall 2024**
- Computational Modeling Tools and Techniques **Winter 2023**
- Introduction to Computational Modeling (MATLAB) **Fall 2022**

Computer Science

- Parallel Computing, Graduate **Fall 2025**
- Parallel Computer Architecture, Graduate **Winter 2025**
- Introduction to Programming II (C++) **Winter 2022**
- Introduction to Programming I (Python) **Fall 2021**

Simulations

- Statistical and Irreversible Thermodynamics, Graduate **Fall 2023**
- Simulation of Condensed Liquids, Graduate **Fall 2023**

Mathematics

- Modern Bayesian Data Science, Graduate **Winter 2024**
- Applied Mathematics for Chemical Engineers, Graduate **Fall 2023**
- Differential Equations **Fall 2020**
- Multivariable Calculus **Winter 2020**

Professional Skills

Coding Proficiencies

- Python (Intermediate – Advanced), C++ (Intermediate), MATLAB (Intermediate), Jupyter (Intermediate-Advanced), Bash (Intermediate)
- High performance computing (Greatlakes, Stampede3, LLNL, LANL), cloud computing (GCS)